

Quiz 1 (Version 1)

CAS CS 132: *Geometric Algorithms*

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- ▷ You will have approximately 35 minutes to complete this exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

1 Row Operations

Apply the row operations:

$$R_3 \leftarrow R_3 + 2R_2$$

$$R_1 \leftrightarrow R_2$$

$$R_3 \leftarrow -2R_3$$

from top to bottom to the following matrix. You must write down the intermediate matrices and row operations you used in your calculation.

$$\begin{bmatrix} 4 & 5 & -1 \\ 2 & -9 & -3 \\ -3 & 5 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 5 & -1 \\ 2 & -9 & -3 \\ -3 & 5 & -2 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + 2R_2} \begin{bmatrix} 4 & 5 & -1 \\ 2 & -9 & -3 \\ 1 & -13 & -8 \end{bmatrix}$$

$$\xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 2 & -9 & -3 \\ 4 & 5 & -1 \\ 1 & -13 & -8 \end{bmatrix}$$

$$\xrightarrow{R_3 \leftarrow -2R_3} \begin{bmatrix} 2 & -9 & -3 \\ 4 & 5 & -1 \\ -2 & 26 & 16 \end{bmatrix}$$

Solution.

$$\begin{bmatrix} 2 & -9 & -3 \\ 4 & 5 & -1 \\ -2 & 26 & 16 \end{bmatrix}$$

2 Reduced Echelon Forms

Determine the reduced echelon form of the following matrix. You must write down the intermediate matrices and row operations you used in your calculation.

$$\begin{bmatrix} 2 & 4 & -6 & 0 \\ -3 & -6 & 9 & 3 \\ -1 & -2 & 3 & -6 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & -6 & 0 \\ -3 & -6 & 9 & 3 \\ -1 & -2 & 3 & -6 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1/2} \begin{bmatrix} 1 & 2 & -3 & 0 \\ -3 & -6 & 9 & 3 \\ -1 & -2 & 3 & -6 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2/3} \begin{bmatrix} 1 & 2 & -3 & 0 \\ 1 & -2 & -3 & 3 \\ -1 & -2 & 3 & -6 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 0 & 0 & 3 \\ -1 & -2 & 3 & -6 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_1} \begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & -6 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 + 2R_1} \begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2/3}$$

Solution.

$$\begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3 General Form Solutions

Determine a general form solution for a linear system whose augmented matrix is row equivalent to the following matrix (*Hint*. First determine if the following matrix is in reduced echelon form).

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 5 & 3 \\ 0 & 1 & 1 & 1 & -1 & 4 \\ 0 & 0 & 1 & 3 & -4 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 5 & 3 \\ 0 & 1 & 0 & -2 & 3 & 0 \\ 0 & 0 & 1 & 3 & -4 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.

$$x_1 = 3 - x_5$$

$$x_2 = 2x_4 - 3x_5$$

$$x_3 = 4 - 3x_4 + 4x_5$$

x_4 is free

x_5 is free